Instance-Derived Seam Supervision for Separating Adjacent Developing Bones in Pediatric Hand Radiographs

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Abstract

Segmenting developing bones in pediatric hand radiographs is difficult when adjacent ossification centers are separated by narrow, low-contrast gaps. Region-overlap objectives can produce high Dice scores while still joining structures that should remain separate. We introduce instance-derived seam supervision, a training-only construction that converts indexed bone annotations into a narrow background class between neighboring instances. The resulting three-class target can be used with a conventional semantic segmentation backbone and requires only the radiograph at inference. We evaluate the construction with matched nnU-Net models on 875 training and 96 validation radiographs under a frozen evaluation protocol. Relative to a mature binary nnU-Net, seam supervision reduces the image-level component under-count rate from 51.04% to 42.71% and the gap-region false-positive rate from 0.1873 to 0.1832. Dice remains nearly unchanged (0.8980 versus 0.8974), while Boundary IoU and 2-pixel Surface Dice increase by 0.0007 and 0.0015. The gains are therefore specific but modest: HD95 and ASSD worsen, revealing occasional boundary outliers. These results show that instance-derived background supervision can reduce adjacent-structure merging, but also motivate confidence weighting and foreground-preservation constraints before the approach is claimed as a general plug-in module.

Introduction

Pediatric hand radiographs contain many small developing bones whose appearance changes substantially with age. In younger children, ossification centers are sparse and often widely separated. As development proceeds, additional centers appear, existing structures enlarge, and some inter-bone gaps become only a few pixels wide. These narrow gaps are easily obscured by low contrast and projection overlap. A segmentation model may therefore achieve strong region overlap while joining two anatomically distinct structures.

This failure is poorly summarized by Dice alone. A bridge that occupies a small number of pixels can merge two components without materially changing the union area. The error is nevertheless conspicuous in a qualitative result and can invalidate downstream measurements based on component count, shape, or spacing. Our goal is not to maximize overlap by expanding or eroding every prediction. Instead, we seek to suppress false foreground specifically inside plausible inter-instance gaps while preserving the bone support on both sides.

The available labels provide a useful but underused signal. They are indexed masks: zero denotes background and each positive value denotes a separate annotated bone instance within an image. We convert these instance identities into a training-only seam target. For each background pixel near the foreground, we identify the nearest annotated instance. A change in nearest-instance identity across neighboring background pixels marks a Voronoi-like separator. This separator is dilated into a narrow seam class, while all bone instances remain a single semantic bone class. At inference, the model receives only the radiograph; no ground-truth seam or instance identifier is supplied.

We test the construction in a controlled comparison between a binary nnU-Net (Isensee et al. 2021) and a matched three-class nnU-Net. The two systems use the same split, patch size, optimization budget, and checkpoint rule. Under the frozen R201 original-validation protocol, seam supervision lowers the image-level component under-count rate by 8.33 percentage points with a Dice change of only -0.0006. Boundary and gap metrics improve slightly, but surface-distance outliers become worse. The evidence therefore supports a narrow claim: explicit seam supervision changes the topology error profile, but the unweighted formulation is not yet a complete solution.

Our contributions are:

- We derive a background seam target directly from indexed instance annotations without modifying the original dataset or using ground truth at inference.
- We introduce a matched evaluation protocol that combines overlap, boundary, surface, gap, and component diagnostics rather than relying on Dice alone.
- We show that mature seam-supervised nnU-Net training reduces component under-counting from 51.04% to 42.71%, while documenting the accompanying recall and surface-distance trade-offs.

The current paper establishes the supervision mechanism and its failure modes on one backbone. A model-agnostic weighted loss and age/sex-conditioned confidence model are

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natural extensions, but their cross-backbone effectiveness remains to be validated.

Related Work

Medical image segmentation backbones. U-Net established the encoder–decoder design with skip connections that remains common in biomedical segmentation (Ronneberger, Fischer, and Brox 2015). nnU-Net subsequently showed that preprocessing, architecture configuration, training, and post-processing choices can matter as much as introducing a specialized network (Isensee et al. 2021). We use nnU-Net as a controlled backbone because it provides a strong semantic baseline. Our contribution is not a new encoder. It changes how instance information supervises background separation.

Boundary-aware learning. Region losses such as cross entropy and Dice aggregate error over areas and may underweight thin interfaces. Boundary loss instead optimizes a contour-distance objective and is particularly relevant to imbalanced structures (Kervadec et al. 2019). Boundary-aware objectives improve localization, but a foreground boundary does not by itself specify whether two nearby foreground regions must remain disconnected. Our seam target focuses on the background channel between annotated instances.

Topology-aware segmentation. Topology-preserving objectives have been developed to maintain connected components, holes, and tubular continuity. Persistent-homology-based losses penalize topological disagreement (Hu et al. 2019), while clDice measures and optimizes centerline agreement for tubular structures (Shit et al. 2021). Discrete Morse theory has also been used to identify critical structures in topology-aware segmentation (Hu et al. 2021). These approaches commonly preserve foreground connectivity. Pediatric bone separation requires a different orientation: the desired property is continuity of the narrow background seam that prevents adjacent foreground instances from merging.

Boundaries in instance separation. Closely packed object and cell segmentation likewise fails when semantic foreground predictions erase inter-instance boundaries. Recent work on cell instance segmentation argues that instance separation depends critically on boundary representation rather than foreground classification alone (Liang et al. 2025). Pediatric hand radiographs share the adjacency problem but introduce different appearance variation: ossification centers change in number, size, and spacing throughout development. We use this analogy to motivate seam supervision while retaining a task-specific evaluation of gap false positives and component under-counting.

Method

Problem Formulation

Let $\mathbf{X} \in \mathbb{R}^{H\times W}$ be a grayscale radiograph and $\mathbf{Y} \in \{0,1,\dots,K\}^{H\times W}$ its indexed annotation. Label zero denotes background, and each positive integer denotes an annotated bone instance within that image. Instance identifiers are local to an image and are not treated as fixed anatomical classes. The semantic bone mask is

$$\mathbf{B}(\mathbf{x}) = \mathbb{1}[\mathbf{Y}(\mathbf{x}) > 0]. \tag{1}$$

The standard model predicts background and bone. Our seam-supervised model additionally predicts a separation class derived from $\mathbf{Y}$ during training.

Instance-Derived Seam Construction

For every background pixel $\mathbf{x}$, we compute its distance to each annotated instance k and assign the nearest identity

$$n(\mathbf{x}) = \operatorname*{arg\,min}_{k\in\{1,\dots,K\}} d(\mathbf{x}, \{\mathbf{z} : \mathbf{Y}(\mathbf{z}) = k\}). \tag{2}$$

A background pixel becomes a seam candidate if a neighboring background pixel has a different nearest identity. We restrict candidates to a radius r from bone foreground:

$$\mathbf{S}(\mathbf{x}) = \mathbb{1}\big[\mathbf{B}(\mathbf{x}) = 0,\ d(\mathbf{x},\mathbf{B}) \le r, \\ \exists \mathbf{q} \in \mathcal{N}(\mathbf{x}) : n(\mathbf{q}) \neq n(\mathbf{x})\big]. \tag{3}$$

We use $r = 12$ pixels and dilate the resulting centerline by one pixel. This construction approximates a local Voronoi separator. It does not assume that an instance index has the same anatomical meaning across subjects.

Three-Class Supervision

The training target $\mathbf{T}$ is

$$\mathbf{T}(\mathbf{x}) = \begin{cases} 1, & \mathbf{B}(\mathbf{x}) = 1, \\ 2, & \mathbf{B}(\mathbf{x}) = 0 \text{ and } \mathbf{S}(\mathbf{x}) = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

The current implementation uses the standard nnU-Net compound cross-entropy and Dice objective over the three classes. During inference, the predicted bone mask contains only class 1; class 2 is excluded as background separation. This version deliberately avoids a hand-tuned seam threshold.

Toward a Decoupled Plug-In Objective

The target construction is independent of the backbone, but the present three-class head is not yet a strict plug-in loss. A decoupled form would retain a binary output probability p_b and impose an upper bound $u(\mathbf{x})$ in high-confidence seam regions:

$$\mathcal{L}_{\text{sep}} = \frac{1}{|\Omega|} \sum_{\mathbf{x}\in\Omega} w(\mathbf{x}) \left[\max(0, p_b(\mathbf{x}) - u(\mathbf{x}))\right]^2. \tag{5}$$

Foreground-support terms would prevent the model from lowering merge errors by indiscriminately eroding bone. Age and sex could condition w and u, but we do not evaluate that extension in the current experiments.

Experiments

Data and Protocol

We evaluate only the TSRS_RSNA-Epiphysis subset. The indexed labels are converted in isolated output directories; the original images and annotations remain unchanged. The

fixed split contains 875 training and 96 original-validation radiographs. A separate clean-test-v2 split is reserved for final evaluation and is not used for threshold selection, checkpoint selection, or the results reported here.

Both systems use a two-dimensional PlainConvUNet configured by nnU-Net, a 512×512 patch, batch size 2, and the same 875/96 split. A 30-epoch matched run is followed by a 150-epoch mature run initialized from the corresponding best weights with a reset optimizer and learning-rate schedule. We select checkpoints using nnU-Net's exponentially smoothed foreground Dice criterion. The control predicts background and bone; the seam model predicts background, bone, and seam.

Metrics

The frozen R201 protocol reports Dice, IoU, precision, recall, Boundary IoU, Boundary F1, Surface Dice at 2 and 5 pixels, HD95, and average symmetric surface distance. Pixel distances are reported because reliable physical spacing is unavailable. We additionally report two task-specific diagnostics. Gap-region false-positive rate measures foreground predictions in a nine-pixel dilated background band around the ground truth. Component merge rate is operationally defined as the fraction of images for which the predicted binary mask contains fewer connected components than the binary ground truth:

$$\mathrm{CMR} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[C(\hat{\mathbf{B}}_i) < C(\mathbf{B}_i)]. \tag{6}$$

It is an image-level component under-count diagnostic, not a universal instance-segmentation metric. Component-count MAE reports $|C(\hat{\mathbf{B}}) - C(\mathbf{B})|$.

Quantitative Results

Table 1 shows a targeted change rather than uniform improvement. CMR falls by 8.33 percentage points, from 51.04% to 42.71%, and gap-region FP falls by 0.0041. Boundary IoU, Boundary F1, and Surface Dice improve slightly. Dice changes by -0.0006, indicating that the component improvement is not explained by a large loss of overlap. However, recall decreases by 0.0025, count MAE increases by 0.104, and HD95 nearly doubles. The seam target therefore suppresses some false bridges but also creates or exposes isolated boundary errors.

Qualitative Analysis

Figure 1 illustrates why overlap and component diagnostics must be reported together. Predictions with similar union area may differ at a narrow bridge. Conversely, a method can remove a bridge by eroding foreground, which motivates the simultaneous recall, surface-distance, and component-count checks.

Discussion and Limitations

The mature experiment answers two questions. First, seam supervision is not merely an early-training artifact: the component under-count improvement persists after matched extended optimization. Second, more training alone does not eliminate the trade-off. The three-class formulation gives a sparse seam class equal status in the objective, but it does not distinguish a reliable inter-instance gap from an uncertain low-contrast boundary. The worse HD95 indicates that a small number of distant errors dominate the surface tail.

The image-level CMR should be interpreted cautiously. It detects component under-counting, but it does not identify which anatomical pair merged, and fragmentation can reduce CMR while worsening component-count MAE. The diagnostic is therefore paired with overlap, recall, boundary, surface, and count metrics. Statistical uncertainty and repeated-seed variation are not yet reported. The dataset also comes from a single pediatric hand-radiograph task, and the age distribution is imbalanced. These limitations prevent a claim of general anatomical topology preservation.

The current implementation uses a three-class head and one backbone. A stronger model-agnostic version should decouple the prior from the architecture: a confidence-weighted seam map would act on ordinary binary logits, and a two-sided support loss would penalize indiscriminate erosion. Age and sex should modulate confidence rather than define a fixed spatial template because hand size, pose, and ossification patterns are not globally aligned. Claims of plug-and-play transfer require matched experiments on at least U-Net, a multiscale decoder, and nnU-Net.

Finally, the seam target is derived from training annotations. The inference path receives only the radiograph, but the distinction must remain explicit to avoid ground-truth leakage. The held-out clean-test-v2 split remains locked until losses, thresholds, and checkpoint rules are frozen.

Conclusion

Instance-derived seam supervision uses information already present in indexed annotations to expose a failure that region overlap largely ignores. On a matched mature nnU-Net comparison, it reduces image-level component under-counting and gap-region false positives with nearly unchanged Dice. The improvement is meaningful but incomplete: recall and surface-distance results show that unweighted seam classification can trade false bridges for local erosion or outliers. The next step is therefore not a stronger claim, but a stricter mechanism—a decoupled confidence-weighted separation loss with foreground support, followed by cross-backbone and demographic subgroup validation.

Table 1: Mature matched comparison on 96 original-validation images. Higher is better for Dice, recall, Boundary IoU/F1, and Surface Dice; lower is better for gap FP, CMR, count MAE, HD95, and ASSD.

Method	Dice	Recall	BIoU	BF1	SD2	SD5	Gap FP	CMR	Count MAE	HD95	ASSD
Binary nnU-Net	**.8980**	**.8991**	.2343	.3731	.6501	.9004	.1873	.5104	**2.333**	**16.35**	**6.17**
Seam supervision	.8974	.8966	**.2350**	**.3742**	**.6515**	**.9008**	**.1832**	**.4271**	2.438	33.39	6.98

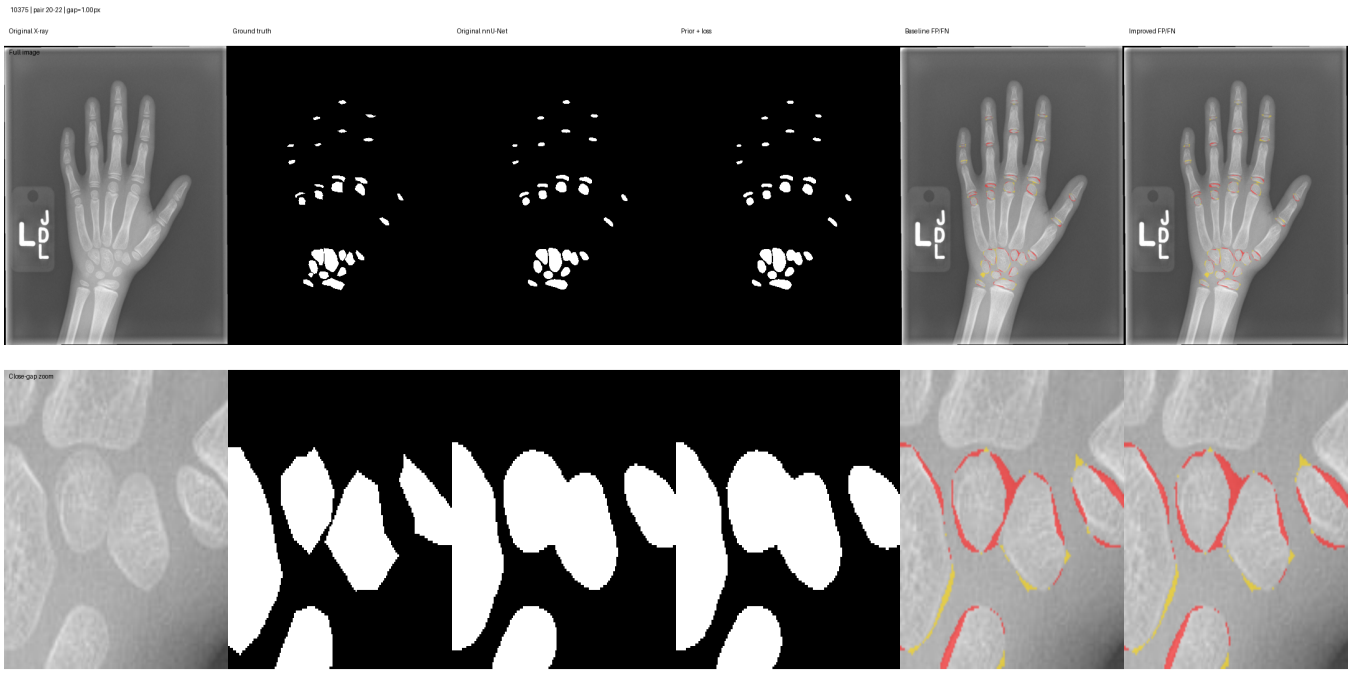


Figure 1: Matched qualitative comparison. The top row shows the full radiograph, label, mature binary nnU-Net, seam-supervised nnU-Net, and error overlays. The bottom row magnifies a one-pixel-gap hard case. Red denotes false-positive foreground and yellow denotes false-negative foreground. The seam model changes the narrow separation geometry, but residual boundary displacement remains.

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Supplementary Implementation Details

The raw indexed label value is treated only as an image-local instance identifier. All positive identifiers are mapped to the semantic bone class. Background Voronoi assignment is computed with a Euclidean distance transform. Seam candidates are restricted to a 12-pixel distance from foreground

and dilated by one pixel. Prediction class 2 is excluded from the final binary bone mask. No ground-truth label, ground-truth seam, or instance identifier is used during inference.

The evaluation uses 3-pixel boundary bands, a 9-pixel gap-region dilation, and Surface Dice tolerances of 2 and 5 pixels. HD95 and ASSD are reported in pixels. All development results use the fixed 96-image original-validation split. The separate clean-test-v2 split is not used in this draft.

Reproducibility Scope

The draft records the exact split size, patch size, batch size, target construction, checkpoint rule, and metric directions. A submission-ready version should additionally report random seeds, repeated-run confidence intervals, age/sex subgroup counts, complete software versions, and the final cross-backbone experiment matrix.